MCSE699J- Dissertation II

A DISTRICT-AWARE MULTI-TASK TEMPORAL INTELLIGENCE SYSTEM FOR CROP SELECTION AND MARKET RISK FORECAST.

Submitted in partial fulfilment of the requirements for the degree of

Master of Technology

in

Computer Science and Engineering (Artificial Intelligence and Machine Learning)

by

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(Deemed to be University under section 3 of UGC Act, 1956)

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DECLARATION

I hereby declare that the dissertation entitled “A DISTRICT-AWARE MULTI-TASK TEMPORAL INTELLIGENCE SYSTEM FOR CROP SELECTION AND MARKET RISK FORECAST.” submitted by me, for the award of the degree of Master of Technology in Computer Science and Engineering (Artificial Intelligence and Machine Learning) to VIT is a record of bonafide work carried out by me under the supervision of Prof. Manjula R, School of Computer Science and Engineering.

I further declare that the work reported in this dissertation has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Vellore **THANYA (24MAI0005)**

Date :

CERTIFICATE

This is to certify that the dissertation entitled "A DISTRICT-AWARE MULTI-TASK TEMPORAL INTELLIGENCE SYSTEM FOR CROP SELECTION AND MARKET RISK FORECAST." submitted by THANYA (24MAI0005), School of Computer Science and Engineering, VIT, for the award of the degree of Master of Technology in Computer Science and Engineering (Artificial Intelligence and Machine Learning), is a record of bonafide work carried out by the student under my supervision during Winter Semester 2025-2026, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The dissertation fulfills the requirements and regulations of the University and in my opinion meets the necessary standards for submission.

Place: Vellore **Signature of the Guide**

Date :

Internal Examiner **External Examiner**

Head of the Department

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EXECUTIVE SUMMARY

Decision-making in agriculture is increasingly influenced by climate variability and market fluctuations, yet a unified predictive framework for crop planning and price forecasting is currently lacking. Most agricultural advisory systems treat crop selection and price forecasting as independent problems. As a result, farmers generally rely on their intuition for crop selection, often failing to match production with market prices. This paper presents a district-aware multi-task temporal intelligence system for unifying crop planning and price forecasting. We integrate time-series historical price data of commodities and crops, crop calendars, meteorological, soil, and yield-related information into a district-aware knowledge base. A Multi-Task Tesentation Network is used to unify crop selection and price-forecasting. A Temporal Convolution (TC) layer, enhanced with attention mechanisms, is utilized to learn temporal information with long-range relations and periodicity. The system learns shared temporal representations and taskspecific embeddings to generate probabilistic price-quantile forecasts, assign volatility classes, and identify suitable crops. By training the model using time-aware data partitioning, improved forecast reliability is achieved over independent tasks. Additionally, quantile regression and cross-entropy loss functions improve the model's ability to predict and risk sensitivity at forward times. The experimental results demonstrate improved crop recommendation and forecast reliability. CropFit forecasts price ranges for crops instead of providing certain values, and thus helps farmers to make economically sound crop selection decisions. It can be used pre-sowing for recommending suitable crops for planting, as well as post-sowing for forward-looking price predictions that guide timely crop sales. CropFit can be seamlessly integrated with existing and emerging decision-support systems for sustainable agriculture and will help bridge the gap between intelligence and practical agricultural application.

Keywords: *Multi-Task Learning, Temporal Convolution Network, Crop Recommendation, Market Price Forecasting, Decision Support System*

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
NLP	Natural Language Processing
QCP	Quantile Coverage Probability
RMSE	Root Mean Squared Error
TCN	Temporal Convolutional Network
TQFN	Temporal Quantile Forecasting Network

SYMBOLS AND NOTATIONS

α_t	Attention weight at time step t
λ_i	Weight assigned to task i in multi-task learning
d	Dilation factor in dilated convolution
e_t	Attention score before normalization
$F(x)$	Learned transformation in residual connection
$H(x)$	Output of residual block
h_t	Hidden representation at time step t
k	Kernel size in convolution
L_i	Loss corresponding to task i
L_q	Quantile loss function
q	Quantile level for probabilistic prediction
R^2	Coefficient of determination
w_i	Convolution filter weights
x_t	Input feature vector at time step t
y_t	Predicted output at time step t
z	Shared temporal feature embedding

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Agriculture remains a vital industry for many economies globally. Farming communities face challenges due to reliance on yearly harvest cycles. Shifts in market demand shape income stability across rural regions. Dependence on timing affects how households manage resources throughout the year. Economic pressure rises when prices change without warning. Livelihood patterns follow climate rhythms rather than long-term planning. Unpredictable conditions define daily life for many agricultural families.

Weather patterns shift more each year, while markets for goods rise and fall without warning. Access to consistent information remains broken across regions. Uncertainty within farming choices has grown, driven by shifts in market insights. Decisions now unfold under heavier doubt, influenced by fluctuating data flows. Producers face these conditions daily, navigating change without clear direction. Crop choices often follow past practice, guided by sight or informal signs of plant health. Still, improved precision may come from structured gathering and application of financial data instead. Yet production fails to meet market demands, causing irregular earnings. Despite more data now exists from the past, With agricultural data, progress in modeling time-based patterns opens a path forward to transition into Productivity tends to rise when methods rooted in science guide preparation. Profit follows such structure, not chance. Where analysis shapes decisions, results improve over time. Efficiency emerges slowly, then clearly. Planning, done with precision, alters outcomes more than expected.

Even though farming data tools have advanced lately, the majority of current tech offerings still fall short in key areas. Among overlooked sections of the decision process, attention often lands on one segment only. Where crop advice tools adjust the farmer's role, expansion goes further elsewhere. Consideration extends to how seeds are shaped by context. Inputs adapt according to surrounding conditions. The setting where choices emerge matters just as much. Shifts happen not through single fixes but layered influences. While many sites about arabica emphasize terrain details or weather fit, predictions of market value rely on deeper analysis using several different factors.

1.1.1 Introduction

Farming stands central to growth in economies and improvement in lives. With countless individuals, earnings and steady access to healthy meals come mainly through crops gathered at specific times each year. Still, shifts in climate trends lately - alongside effects felt across farming - and unpredictable international pricing now cloud choices about what to plant. Decisions among smallholders often follow tradition, memory of earlier seasons, or advice passed from neighbor to neighbor. Yet such methods may overlook changing needs, whether close by or far away, shaped by shifting buyer interests.

Suddenly, tools for gathering information along with techniques for processing it have grown stronger across numerous fields. Not even farming, often seen as straightforward, remains untouched by these shifts. Information about past crops - including climate patterns, ground conditions, soil details, and pricing trends - now exists in usable forms. Systems built with intelligence draw from time-stamped records to guide choices in growing cycles. When models learn from sequences over time, outcomes improve: harvests gain strength while expenses drop gradually behind the scenes.

1.1.2 Problem Statement

Despite accessible tools, today's farming guidance platforms fall short when aiding choices during critical stages of growing and selling crops. While some models analyze soil data alongside climate patterns, others focus solely on market behavior after harvest. Linking what farmers plant to anticipated prices remains overlooked by nearly all current approaches. Rarely does a system merge these elements meaningfully.

Despite offering data, most modern setups fail to link it with local farming conditions or regional needs. Instead of forecasts, they deliver fixed advice that does not evolve. Because of this, guidance often misses timing, location, or economic shifts. Farmers then struggle to pick suitable crops aligned with demand patterns. Incomes rise and fall unpredictably under such uncertainty. Resources remain underused when insights lack motion or adaptation.

A solution must combine crop advice with market predictions while accounting for timing differences across areas. Such a structure adjusts outputs based on location-specific patterns instead of general trends. Time factors shape outcomes just as much as regional conditions do within this framework. Forecasts emerge from layered inputs rather than single-point data sources. Each recommendation reflects shifts occurring at different stages throughout the cycle.

1.2 MOTIVATION

This section explains the driving factors behind undertaking this project.

- Agricultural inefficiencies lead to unstable incomes and food supply issues, making improved decision-making essential.
- Farmers need reliable predictive tools to make better crop selection and market planning decisions.
- There is a research gap in integrating crop recommendation and price forecasting into a unified system.
- The project explores advanced machine learning techniques for handling complex temporal agricultural data.

1.3 SCOPE OF THE PROJECT

This section defines the boundaries and limitations of the project.

- The system provides crop recommendations, predicts future prices, estimates market risks, and supports decision-making across pre-sowing and post-sowing stages.
- The system ensures scalability, reliability, usability, performance efficiency, and data security.
- The project utilizes machine learning and deep learning techniques with multi-source agricultural data for prediction and analysis.
- The system operates under assumptions of data availability and quality while being constrained by data limitations, environmental uncertainty, computational resources, and deployment challenges.

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

2.1 LITERATURE REVIEW

Agricultural intelligence using machine learning and deep learning techniques has attracted a lot of attention in recent years, especially for crop planning and market forecasting. This attention has led to a variety of approaches that combine environmental v, historical, and predictive analytics in order to support intelligent decision making in agriculture. Most of these studies have focused on particular tasks such as crop recommendation, yield prediction, and price forecasting. However, there is a lack of unified approaches for solving these tasks in order to support comprehensive decision making in agriculture.

Several studies have explored ensemble of different supervised learning approaches for forming crop recommendation systems. Building upon such idea, Bhavya Sree, Jyoti Rastogi, Pooja Badola, Srishti Ojha, Pooja Shukla and Pankaj Katara in their paperJunankar et al. (2023) integrated various learning approaches like Support Vector Machine, Random Forest, Gradient Boosting Regression and more to develop and deploy a crop-recommendation web platform-AgriFusion, for selecting most suitable crop on the basis of climatic and soil parameters. They utilized datasets like those available on Kaggle platform related to different crops such as their characteristics and crop-yield. The employed models were trained and validated using a large number of instances and their predictive accuracy was compared. The system has advantage of ensemble approach which makes it robust and less prone to bias; however, it fails to consider future market trends which are very important for taking any economic decision related to crop.

In like manner, Menaka and colleaguesKrishna et al. (2025) introduced a forecasting system using Random Forest together with Gradient Boosting to estimate harvest output, whereas ARIMA served long-range cost projections. The dataset included indicators of soil quality, atmospheric patterns, past productivity figures, along with pricing trends, allowing algorithms to detect complex farming dependencies. Findings show ensemble techniques handle extensive farm-related datasets well, enhancing forecast accuracy. Even so, the research handles volume forecasts and value estimates independently, without linking outcomes to practical planting decisions.

Looking beyond standard methods, Thakre et al.et al. (2026a) explore how machine learning handles complex data to anticipate shifts in commodity prices. Instead of relying on traditional statistics, which often fall short when patterns change rapidly, their study favors supervised approaches for better results. Through adjustments like refining inputs, reducing variables, and scaling values, system output improves noticeably. Although the proposed method offers useful guidance for building adaptable forecasting systems, issues around local relevance and live deployment persist. With that, room remains to develop models more responsive to specific environmental conditions.

Further progress in farming forecasts comes through deep learning systems using location-based data. Instead of traditional methods, a mix of Graph Neural Networks and Convolutional Neural Networks was applied by Bhardwaj et al.Chibuye et al. (2025), capturing links between distant farm markets. Public records on tomato and potato prices helped test the system, showing stronger results than earlier models while predicting trends as far as one month forward. Because nearby regions often influence each other, accounting for distance improved forecast stability. Despite these gains, the work centers only on pricing, leaving out planting strategies, which reduces usefulness when broader choices are needed.

Alongside these advances, work by Aher et al. et al. (2022) introduced a Market Intelligence System combining ARIMA models for immediate predictions alongside Long Short-Term Memory networks to detect prolonged nonlinear trends. Instead of relying solely on static inputs, it uses both current and past information to generate practical guidance suitable for growers and decision-makers, its structure scaling efficiently due to an interface built around user needs. Such blended methods have proven capable when simulating unpredictable farm market behavior. Even so, the analysis points out that numerous smart frameworks remain unable to operate fully once deployed, typically struggling to merge ongoing data flows with accessible interfaces tailored to agricultural users - revealing a disconnect between experimental designs and field-ready solutions.

A method presented by Junankar et al.et al. (2026b), known as the Temporal Fusion Transformer, was applied to forecast wheat yields through an advanced attention mechanism. Despite strong accuracy and clarity in identifying key variables, concerns remain about reliance on complex models. Fueled by diverse datasets from top wheat-growing regions, performance exceeded traditional machine learning techniques. Because timing patterns matter, capturing sequences becomes essential - this framework does so while revealing how each factor contributes. Yet complexity often breeds skepticism; such systems are frequently seen as opaque. In farming contexts where understanding decisions matters, opacity reduces trust. This limitation is noted directly within the analysis.Chibuye et al. (2025)

Using TFT beyond its original scope, Krishna et al.et al. (2026c) designed an irrigation setup responsive to weather conditions by combining IoT sensors with multivariate forecasting over time. Instead of relying on isolated measurements, the approach analyzes patterns in soil moisture, temperature, humidity, and rain inputs to determine when watering is needed. When tested, it delivered better forecasts than standard methods like LSTM and GRU while using fewer resources. Through examining which variables mattered most, clarity about decision logic increased, aiding trust in automated choices. Despite progress, difficulties persist - sensor adjustments, mismatched device types, varying crop needs, and regional differences limit broad usage. As a result, widespread adoption faces unresolved hurdles.Sree et al. (2024)

Data at scale helps clarify how farming choices unfold in practice. From Central Europe, more than 16 million field entries were examined by Palka et al.Junankar et al. (2023), applying a random forest method to detect what steers crop sequences. Prior crops grown, advice on farm practices, weather factors, market values, along with policy frameworks - each plays a role, shaping decisions together. Such insights form among the broadest collections of evidence available for managing crop operations, showing clearly that forecasts must account for unpredictability in grower actions. Even so, established planning methods tend to depend heavily on fixed agronomic guidelines, which rarely match actual variation seen on ground. This gap signals demand for modeling approaches responsive enough to represent complex, evolving influences behind land-use choices.

From 2010 onward, Chibuye and team Krishna et al. (2025) examined how Fall Armyworm impacts maize harvests using a dynamic Echo State Network tailored for prediction tasks. Instead of treating pests as separate factors, their system blends on satellite derived greenness measures, climate records, earth composition details, alongside insect monitoring figures across many growing cycles. Because it processes these layers together, forecast inaccuracies dropped noticeably especially when invasions grew intense. What sets this method apart begins with a built-in adjustment rule: higher pest levels trigger automatic corrections in output estimates, making outcomes align more closely with actual field damage. While earlier systems ignored such living threats entirely, this version runs efficiently enough for live use yet exposes weaknesses found in standard crop modeling practices where biology often gets left out.

Beyond predictive modeling, work by Pasut et al. Palka et al. (2026) examined how soil monitoring information, when combined with machine learning methods, could chart extended patterns of tillage and residue handling in Australian agriculture. Drawing upon over twenty years of field-level observations, random forest techniques detected shifts across space and time, outperforming both logistic regression and nearest-neighbor approaches in precision. Insights into predictor relevance revealed that timing elements and geographic traits strongly shaped practice uptake. Still, challenges remain due to gaps in recorded farming activities alongside constraints inherent in broad national models, which often mask fine-grained variations - pointing toward a need for tailored regional forecasting systems.Bhardwaj et al. (2025)

Beyond earlier techniques, Betew et al.Palka et al. (2026) examined shifts in crop yield forecasting, moving from basic statistical regressions toward intricate neural networks. Performance gains appear evident - recent systems show R^2 values from 0.85 up to 0.93, whereas older ones fall within 0.60 to 0.75. While combining diverse data types improves outcomes, verification by domain specialists remains essential for credibility. Despite these advances, high computing needs along with challenges in areas lacking sufficient data suggest a need for careful trade-offs between precision and reach. Although powerful, newer frameworks are not universally applicable under current constraints.

Learning based on graphs now plays a key role in handling intricate relationships among data points. Rather than sticking to standard methods, Gong et al.Pasut et al. (2026) introduced EXP-GNN, a system for finishing knowledge graphs by refining how information moves across nodes. Stronger movement rules, it turns out, allow better capture of nearby node traits - this insight comes with mathematical backing. As shown in tests, using unique messaging and combining steps boosts model performance noticeably. Still, much current work in graph networks leans on trial-and-error instead of clear design logic. That tendency highlights a gap where systematic frameworks could offer clearer direction.

Though focused on scale limits in graph networks, Jiang et al. Pasut et al. (2026) examined how sampling methods shape training across massive graphs. Rather than rely solely on faster hardware, their analysis separates fixes into four types: equipment upgrades, leaner models, data selection tactics, and combined designs. Iterative updates during learning tend to lift results more noticeably when compared to adjustments in sample structure. One method - extracting smaller subgraphs - cuts processing load without weakening accuracy, fitting well within extensive data settings. As graphs grow larger, memory demands rise alongside training difficulty, a pattern highlighting why smart design choices matter. Efficiency must align with prediction quality, especially under expanding conditions Thakre et al. (2025)

Feature integration from diverse data types improves prediction performance in multi-view learning. Introduced by Wu et al. (2026), a method combining spatial and spectral traits uses structured sparse canonical correlation for identifying depression. This approach applies constraints promoting sparsity and continuity, lowering the interference without disrupting collective patterns - accuracy reaches above 86%. While fusion techniques rooted in deep networks show flexibility, issues around intricacy, transparency, and extensive data needs arise. Such demands might limit real-world implementation where computational capacity is limited. et al. (2026a)

Rather than stand apart, the work by Fortin et al. et al. (2026d) aligns with advances in predictive modeling through a Web API-driven method for generating weather data. With this structure, climate inputs flow via server responses, allowing models to access information without local storage. Independent systems now link more smoothly, thanks to standardized communication patterns replacing isolated processes. Over time, self-contained generators have demanded repetitive handling - downloading, formatting, checking. Yet when integrated into broader workflows, these components sometimes clash with host environments. Updates grow harder as dependencies multiply across layers. Modular setups begin to matter greatly under such conditions, enabling growth without fragmentation S and C (2024).

A method by Luo et al. uses enhanced graph representations within a temporal convolutional setup to better detect anomalies when data gaps exist Luo et al. (2026). Instead of combining approaches directly, it links reconstruction and forecasting tasks through a masking-aware graph attention process that assesses reliability loss from incomplete inputs. Performance gains appear consistently across diverse test cases, especially where faults emerge sooner than expected. In another case, work led by Wang et al. applies layered temporal filters together with blended attention modules to estimate how long aircraft engines will remain functional before maintenance is due Wang et al. (2026b). This approach applies constraints promoting sparsity and continuity, lowering interference without disrupting collective patterns - accuracy reaches above 86%. While fusion techniques rooted in deep networks show flexibility, issues around intricacy, transparency, and extensive data needs arise. While these methods advance sequence understanding, they do not extend naturally to wider systems needing combined predictions and guidance outputs.

Later on, methods rooted in time-focused convolutions began influencing how the space-related patterns are analyzed alongside mixed forecasting approaches. Instead of treating locations separately, Kannadasan and Yogeswari built a system grounded in geographic data access principles, applying a flexible convolution structure adjusted via rule-guided selection of settings Kannadasan and Yogeswari (2026). This approach is handles both location and timing relationships across transit systems Wang et al. (2026a). This method shows broad applicability due to combined graph and hypergraph structures, although narrow focus restricts adaptation across fields. In parallel, work by Mu et al. introduces a mix of LSTM and enhanced temporal convolution for predicting immediate power needs, using maximum mutual information for trait filtering along with self-attention layers Mu et al. (2025). Prediction precision improves notably under their design; still, emphasis remains on electricity usage patterns rather than decisions shaped by context.

Together, such research highlights a shift toward predictive models that integrate data, scale effectively, and offer clarity within farming and ecological settings. Accuracy in forecasts has increased thanks to advances like deep learning, and combining multiple data perspectives - yet several current approaches still struggle with high computing demands, difficulties merging information sourcesAher et al. (2025)

2.2 RESEARCH GAP

Based on the literature survey conducted, below the the identified research gaps.

- Crop suitability and market price forecasting are studied separately
- Limitations in previous studies
- Price models rarely consider district-level crop context
- Most systems give point forecasts, not price ranges or risk
- No unified system supports both pre-sowing and post-sowing decisions

2.3 OBJECTIVES

This work seeks to build a time-sensitive system aware of regional differences, designed to guide choices throughout growing seasons. Instead, it merges forecast methods with evaluation of uncertainty to refine planting strategies alongside readiness for trade shifts through diverse farm-related datasets. Through balancing model accuracy against real-world use, the effort intends to offer a method adaptable at scale - supporting income resilience while advancing agriculture shaped by evidence. Ultimately, such tools may shift how field-level decisions align with broader economic patterns.

- To develop a district-level agricultural dataset that integrates historical prices, crop calendars, environmental indicators, and yield-related attributes for unified temporal analysis.
- To design a multi-task temporal deep learning architecture capable of jointly predicting crop suitability, price quantiles, and market volatility.
- To evaluate the proposed model against baseline forecasting approaches using statistical accuracy metrics and decision-level economic outcomes.
- To enhance risk-aware agricultural planning by generating probabilistic price ranges instead of single-point forecasts.
- To implement a deployable decision-support framework that assists farmers in both pre-sowing crop selection and post-sowing market strategy.

2.4 PROBLEM STATEMENT

Agriculture will continue to face challenges when the systems remain unlinked. In the absence of cohesion, coordination will be compromised across the key operations. The fragmented approaches only delay progress in the resource allocation. The disconnected data streams hamper the making of the right decisions at the right time. Progress gets hampered when different components fail to interact at a more convenient level. Efficiency goes down when management models are disjointed. Usually, coherence hardly happens unless there is a deliberate structural alignment.

- Absence of a unified system that jointly supports crop recommendation and market price forecasting.
- Limited incorporation of district-level context in predictive agricultural models.
- Dependence on single-point price predictions that do not capture market uncertainty or volatility.
- Lack of predictive decision-support tools that assist farmers in both pre-sowing planning and post-sowing market strategy.

2.5 PROJECT PLAN

This project targets to build a district-aware agricultural decision support system that incorporates multi-source data such as historical price, weather, field characteristics and crop yield data. The system processes and prepares the input data, followed by a series of feature engineering techniques. A Temporal Quantile Forecasting Network with convolution and attention mechanism will be developed for district-aware price forecasting. It further leverages environmental and economic factors in a hybrid crop recommendation module, which generates top-k risk-aware crop suggestions for farmers. The system will be developed in an end-to-end pipeline and is capable of making pre-sowing and post-sowing decisions. Comprehensive evaluation will be performed by measuring accuracy, error and applicability on real-world scenarios to ensure system's scalability.

CHAPTER 3

TECHNICAL SPECIFICATION

3.1 REQUIREMENTS

3.1.1 Hardware Configuration

Listed below are the hardware configuration required for this project to be achieved.

- **Processor** – AMD Ryzen 7 7840HS processor (8 cores, 16 threads, 3.8 GHz base clock speed, up to 5.1 GHz turbo boost).
- **GPU** – NVIDIA RTX 4050 Laptop GPU with 6 GB GDDR6 VRAM (96-bit memory interface).
- **Memory (RAM)** – 16 GB DDR5 (4800 MHz).
- **Storage** –1 TB NVMe SSD
- **Power Consumption** – 120 W TDP optimized for model fine-tuning.
- **Cooling System** – Dual-fan liquid-cooled thermal management for stable GPU training temperatures (¡ 75 °C).

3.1.2 Software Configuration

Listed below are the Software configuration required for this project to be achieved.

- **Operating System** – Windows 11 Pro (64-bit).
- **Python Version** – Python 3.10.13.
- **PyTorch** – Version 2.3.1 (CUDA 12.1 enabled).
- **Transformers (Hugging Face)** – Version 4.44.0.
- **Accelerate** – Version 0.33.0.
- **BitsAndBytes** – Version 0.43.1 (for 4-bit quantization support).
- **Tokenizers** – Version 0.19.1.

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

This work presents an overview of the architecture of our end-to-end system that processes large amounts of heterogeneous agricultural data and turns it into decision intelligence. Figure 4.1 highlights the various steps of the processing pipeline that starts from collecting historical or streaming data from various external data sources (e.g., historical prices, weather, sensors, crowdsourcing), continues with data preprocessing in order to correct, complete and align in time the processed information, and ends with feature engineering by creating a temporal and spatial set of features that reflect the variability of agricultural outputs and inputs.

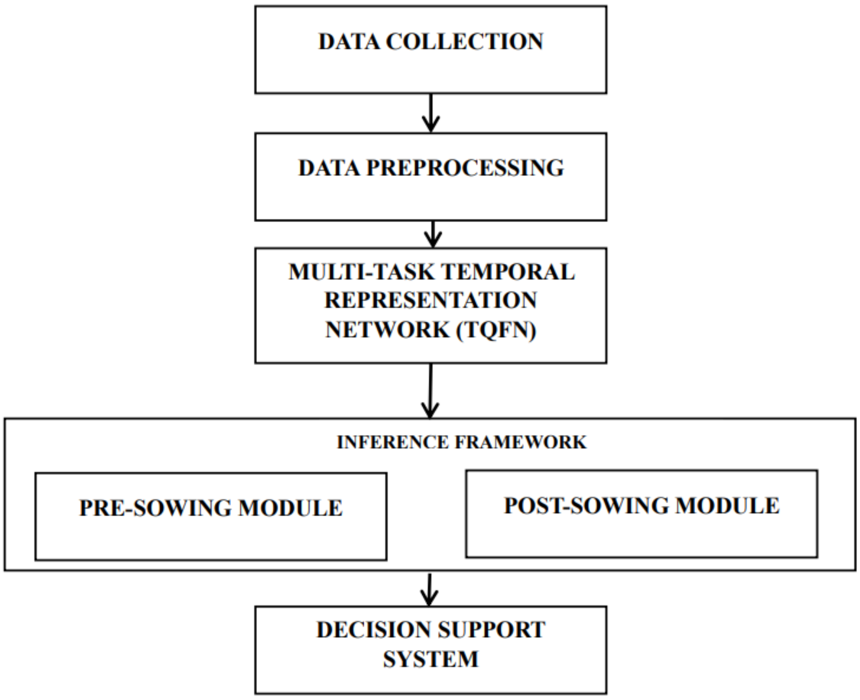

Figure 4.1: Block diagram of model architecture.

In the Figure 4.1 , we show how the engineered features are fed into the Multi-Task Temporal Representation Network, which serves as the core of the architecture and learns to improve performance on a suite of tasks during training. The trained models are then incorporated into a decision support system that provides farmers with the most suitable crop, price forecast and volatility for their area of land.

The architecture in Figure 4.1 consists of separate pipelines, each consisting of several steps that can be developed independently (layers). This modularity improves scalability of the overall architecture. In addition, all steps can follow a data trace, making predictions interpretable down to the level of the input data. The pipelstem (pipeline + stem) for balancedInput addresses a general problem in existing agricultural advisory services and improves decision-support for farm management through data-driven and connected methods.

4.2 DETAILED DESIGN

4.2.1 Dataset Description

One foundation of the design lies in an agricultural reporting network operating at the district scale, drawing on past records of harvest volumes and trading outcomes. From multiple channels, both digital and physical, earlier patterns in crop pricing were gathered - specifically through visits to the Agmarknet portal, delivering day-by-day valuations across regions over extended durations. Instead of relying solely on numbers, scheduling guides for planting cycles appear alongside maps matching crops to districts, revealing what grows where and when choices tend to cluster. Alongside these layers, details tied to growing conditions - including rainfall, earth composition, and output per hectare - are fed into the structure. What emerges includes not just trade trends but underlying natural influences shaping them.

Numbers recorded over time make up the core of the dataset, alongside labels for location and produce kind. Though limited in regional scope, the focus on specific areas supports reliable patterns across seasons. From one season to another, shifts in pricing emerge clearly when viewed through localized trends. Patterns tied to geography become visible because each area contributes distinct agricultural traits. Over months, price behaviors adapt in ways linked closely to where crops grow. Not every region appears here; choices were made to keep comparisons meaningful. Instead of broad claims, insights come from observing how places differ year after year. Temporal movement in values connects with land-specific conditions naturally. Only certain crops appear, chosen so that changes over time stay measurable. Where farming differs, so do economic rhythms, captured point by point. Consistency shapes selection - without it, variation would obscure real signals. As years pass, repeated cycles offer insight shaped by place and harvest type. Realism guides inclusion: what remains reflects actual growing conditions. Each entry stands as a marker between timing, terrain, and trade. Predictive strength grows not from volume but from careful alignment of factors.

4.2.2 Data Processing Module

We used a step-by-step method to develop a preprocessing pipeline that would transform the data into a form that is suitable for time-sensitive analysis. As the first step, we removed duplicate records; then, for filling in missing values (training purposes only), we used different functions; we also divided the data according to location (spatial consistency); finally, we aligned the time intervals during which sequences were labeled (time-base sequence analysis consistency).

Training can be made more stable by scaling and normalizing the data, and also appropriate encoding of categorical data (crop and regions) could be done so that the embedded structures in time are meaningful. Just plain time sequences did not work; after using delayed values (a time window of 10), training the model was able to detect the immediate patterns as well as the long-term patterns. Window average of prices in certain time duration could indicate the direction of the market and the fluctuation of that can indicate uncertainty in that. Also, appropriate seasonal markers can reveal the periodic nature of the farming cycle.

Post preprocessing, there were no inconsistencies in the data points across time and the data was analysis-ready and it could be used for simultaneous multi-task learning also.

4.2.3 Model Architecture

Starting with the raw farm-related information flows into the system through varied sources, shaping the initial stage of processing. After gathering, it gets cleaned and adjusted so inconsistencies fade away quietly. Next comes pattern spotting - time-based rhythms and location-linked trends emerge without fanfare. Into the network they go, a model built for handling several goals at once using common underlying structures. Learning happens in layers, connections form subtly, knowledge spreads behind the scenes. Predictions pop up tailored to specific needs, shaped by what came before. Outcomes arrive on screen through tools designed to guide choices about when to plant crops or weigh economic uncertainties later. From messy numbers to clear outcomes, the flow stays steady no matter which region feeds it. Consistency holds even as details shift beneath

Built around actual farm cycles, this setup handles two key phases. Before the planting begins, predictions about market prices appear alongside warnings about swings in value, then suggestions come through for which crops might earn the most. Once seeds are in the ground, focus shifts - price ranges stretch out ahead, patterns start forming, choices about when to sell gain clearer backing. One unified structure ties it together, cutting the usual split between advice tools found across farming today. Decisions stand firmer because of that link.

Built around a common timeline, the system supports several forecast goals at once, linking farming-related predictions through shared timing patterns. From cleaned inputs, information moves into a sequence handler that tracks time-based changes, then splits toward distinct endpoints - predicting prices here, labeling market swings there, judging crop fit somewhere else. A sketch of the setup shows how data shaping, time-aware processing, learning phases, and advisory functions connect step by step. Splitting roles this way makes growth easier, opening doors later for new climate inputs or local data sources without rebuilding everything.

Temporal Convolution Network Architecture

In our approach the Temporal Convolution Network (TCN) is used as the core part of the model, illustrated in Fig. 4.2. To generate proper future forecasts in a chronological order, we utilize causal (LeakY) Convolations where every element in a network is influenced only by previous elements. A causal convolution operation is denoted by:

$$y_t = \sum_{i=0}^{k-1} w_i \cdot x_{t-i} \tag{4.1}$$

Although expanding the receptive field can help capture both long-term seasonal information and short-term price movements, it significantly increases computational cost. To address this problem, dilated convolution is utilized in this work, where the receptive field is expanded without a huge increase in computational cost.

$$y_t = \sum_{i=0}^{k-1} w_i \cdot x_{t-d\cdot i} \tag{4.2}$$

In order to train deeper networks, residual connections were introduced to maintain stable training. A residual block can be defined as:

$$H(x) = F(x) + x \tag{4.3}$$

An attention mechanism is stacked on top of the convolutional blocks to look at the most relevant time steps in the past and focus on most relevant features for a good prediction in the future. This future-looking ability enables the network to identify sudden market events, periodic harvest events, as well as region specific behaviors.

Out of all the time-based features, some get reused by different tasks, linking their insights without losing individual purpose. Because of this setup, the system handles diverse regions and crop types more effectively. You could write that common structure like this:

$$z = \phi(X) \tag{4.4}$$

where $\phi(\cdot)$ represents the feature extraction function.

Because it mixes time-based filters with smart feature choices, predictions get more accurate. Instead of just one guess, showing several possible results helps cut down on doubt - especially when prices are involved. With stretched-out convolutions, patterns across different time spans come into view, something simpler models usually miss. To measure how well those ranges perform, there's a specific scoring rule based on percentiles

$$L_q(y, \hat{y}) = \max(q(y - \hat{y}), (q - 1)(y - \hat{y})) \tag{4.5}$$

At the same time, training on several tasks helps the model avoid overfitting by nudging it toward patterns that work well across different goals. The multi-task loss can be expressed as:

$$L = \sum_{i=1}^{T} \lambda_i L_i \tag{4.6}$$

where L_i represents the loss for task i and λ_i are task-specific weights.

Later moments in data stay out of earlier model views, making predictions less likely to mislead. Instead of memorizing noise, the setup uses tricks like randomly skipping units or halting before excess tweaks take hold. Each layer's role shifts slightly under such rules, nudging results toward real-world behavior. What grows from this mix is a sharper eye on what crops might thrive where - tied not just to soil but to coming prices. Decisions transformed into code start less rigid when shaped by place and timing alike.

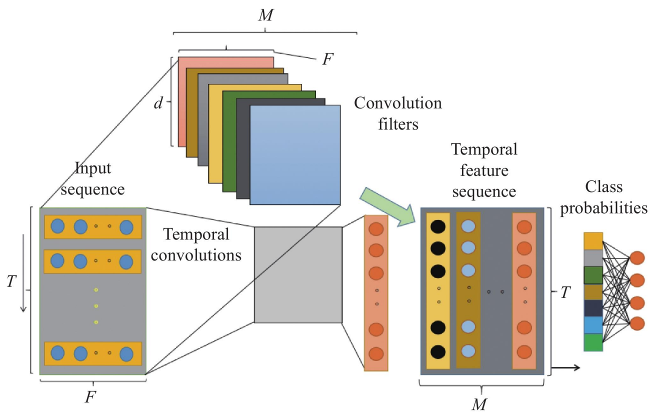

Figure 4.2: Temporal Convolutional Network (TCN) architecture

Figure 4.2 shows how a Temporal Convolutional Network (TCN) is built for handling sequences. Starting from the left, a sequence comes in - length T, with F features per step - and moves into several layers where time-based filtering happens. Instead of piling on more layers, dilation (shown as d) stretches each filter's reach across steps; that way, nearby and distant links in time get noticed, all while keeping computation lean. As data flows forward, these layered filters pull out increasingly structured timing clues, shaping them into a sharper version of the original timeline. A string of values goes next into a dense layer, which turns patterns into likelihoods for each category - showing what the model finally decides. From start to finish, the design shows how TCN reshapes time-based inputs into sensible outputs using spaced-out filters along with layered insight.

CHAPTER 5

METHODOLOGY AND TESTING

5.1 METHODOLOGY

Temporal Quantile Forecasting Network outperforms both older methods and newer ones in meaningful ways. Instead of relying on rigid structures, it tackles core issues like time-sensitive patterns, handling unpredictability, and supporting real choices. Take ARIMA or SARIMA - common tools - they assume straight-line trends, failing when farm prices swing wildly in non-straightforward ways. Meanwhile, models built on regression ignore timing cues baked into data streams. Even advanced options like LSTM or BiLSTM struggle despite their skill with sequences; slow step-by-step processing holds them back, along with fading signal strength across long chains. Now here's something different - the TCN setup we use teams up with attention, letting it handle sequences in parallel while keeping gradients steady and spotting quick patterns alongside slow ones. Instead of locking onto single guesses like most systems do, ours sketches out a range of possible outcomes, showing uncertainty through clear ranges you can actually follow.

Starting off differently, the suggested approach helps people make choices that pay attention to risks while keeping money matters in check - this really matters when farming's involved. Instead of giving just a single number, the system shows several price options, so those using it can plan around optimistic, typical, or tough outcomes without guessing blindly. What stands out is how smoothly it fits into live operations, mainly because it runs fast enough even under pressure.

5.1.1 Loss Function and Optimization Strategy

Starting from probability theory, the approach uses quantile regression to handle unpredictability in crop market pricing. Rather than delivering one fixed result like standard methods, it outputs three distinct quantiles - Q10, Q50, and Q90 - that reflect varying degrees of certainty. Because of this structure, forecasted prices include a range: a floor, a middle value, and a ceiling. With these bounds available, choices can be made while accounting for possible price swings.

Each quantile output is trained using the pinball loss function, which is defined as:

$$\mathscr{L}_\tau(y,\hat{y}) = \max\left(\tau(y-\hat{y}),(\tau-1)(y-\hat{y})\right) \tag{5.1}$$

For a given quantile level τ, the pair $(y,\hat{y})$ stands for observed and forecasted outcomes. Because the penalty changes based on whether predictions fall short or go too far, learning shifts toward capturing how price distributions respond conditionally. While errors below the target face one weight, those above follow another path - shaping model behavior through imbalance.

One benefit of using separate heads for quantile forecasting is how they maintain the unique patterns seen in farm product prices. Because each head trains on its own target, the model captures varied outcomes without interference. Loss values from every head come together into a single score, guiding updates through training. This setup allows several uncertainty ranges to improve at once during learning.

Because gradients guide parameter updates, training moves quickly through complex data landscapes. When quantile loss meets long-term patterns in time-series learning, forecasts gain both precision and confidence ranges - useful when crop prices swing unpredictably. Risk awareness becomes embedded, not added after.

5.1.2 Training Strategy and Regularization Techniques

Beginning with earlier timestamps, the model learns progressively through ordered data segments. As time moves forward, validation and test sets follow training periods without overlap. Later points stay hidden during each training phase, blocking access to future inputs. Sequence structure remains intact throughout partitioning steps. Evaluation happens strictly on samples that come after learned patterns. Learning unfolds step by step, aligned with real-world timing. Future moments never influence present updates. Order preservation supports realistic prediction conditions. Performance reflects how well past instances inform what comes next.

Starting with diverse inputs - past price trends, atmospheric patterns, ground composition, and harvest records - the dataset takes shape through careful cleaning and time-based synchronization prior to model use. To sharpen predictive potential, methods like delayed value markers, moving averages, and time-pattern coding weave into the structure, shaping a richer context for the algorithm to learn from.

One way to help models perform better on new data is through methods that limit overfitting. Instead of relying too much on certain neuron activations, some units get temporarily switched off during learning. This happens inside dropout layers, which break up rigid patterns by chance. When progress on held-out examples stalls, training ends early - avoiding unnecessary updates. Another factor comes from how layers process sequences: convolutions across time steps keep gradients more stable. Unlike loops found in older designs, these structures sidestep fading signal problems.

Though batch-wise training helps handle massive datasets smoothly, updates to model weights happen step by step using backpropagation. Because the method splits data with attention to timing, uses detailed features, and applies constraints during learning, results stay consistent even under varying farm environments. Thanks to these choices in setup, the system adapts well to distinct crops and regions without losing precision or strength in predictions.

5.2 DECISION SUPPORT AND INFERENCE FRAMEWORK

What runs underneath the whole setup? That piece takes raw results from the model and turns them into practical farming guidance using one clear sequence of steps. Built in separate blocks, it links directly to the forecasting network after training, handling details like what crop, where, and when without hiccups. Pulling past data, weather factors, and local conditions on demand keeps each forecast tied closely to real-world cycles and farm-level facts.

From forecasts to fieldwork, this tool connects number-crunching models with actual farming choices through layered reasoning built into its flow. Instead of handing out raw results, it shapes them around realistic conditions so people see what really matters when weighing risks. Built in two clear phases, the process delivers useful guidance ahead of planting time and again afterward, matching timing needs on the ground. With insight tuned to those moments, trust grows along with better outcomes across seasons.

5.2.1 Pre-Sowing Decision Support (Crop Recommendation Phase)

Midway through planning before seeds go into the ground, the system lines up crop choices using expected prices alongside local conditions. Starting from soil type, it folds in climate outlooks along with past harvest records to weigh which plants fit best where and when. Instead of guessing, it blends field data with economic projections so suggestions make sense on farms and at markets.

One score pulls together different checks for every crop. Profit comes from middle-price forecasts instead of averages. To measure danger, they look at gaps across forecast levels - wider space means more doubt in pricing. Soil quality and weather fit help judge if the land likes a certain plant. From that mix, crops get lined up by how well they might do - with odds attached per option - giving farmers clearer paths forward.

5.2.2 Post-Sowing Decision Support (Market Intelligence Phase)

After planting, attention turns to market insights through likely price estimates for the chosen crop. Rather than giving one fixed number, the tool delivers a spread of potential values based on quantile forecasting. Users see how prices might stretch across high, typical, and low ends. Seeing these possibilities helps handle unpredictability in farm economics.

Sometimes guesses about prices help plan when to gather crops, keep them safe, then sell. Looking ahead at likely market shifts lets growers pick better moments to move grain, avoid waste, yet gain more value. Smart forecasts mixed with real farming logic turn numbers into steps people can actually follow. Working well before planting and after it happens makes the whole approach stronger, turning foresight into clear choices in fields.

CHAPTER 6
PROJECT IMPLEMENTATION

6.1 SYSTEM DEVELOPMENT AND DATA PROCESSING

Starting off, various farm-related information gets pulled together - market numbers from past years, local weather records, details about dirt makeup, alongside harvest results - to reflect what really shapes how crops are managed. Because raw inputs often arrive messy, they go through fixes like clearing errors, syncing timelines across sources, smoothing out gaps, plus tossing duplicate entries so everything lines up properly later on. To pull useful clues from trends over time, methods reshape the facts: prior period snapshots emerge, moving averages form, yearly cycles get coded, while group labels turn into number forms behind the scenes. Once cleaned and expanded, all pieces join into one complete picture divided carefully by date order - not shuffled - so earlier moments train models, middle parts guide tuning, final segments check performance without peeking ahead.

6.2 MODEL IMPLEMENTATION AND DEPLOYMENT

Starting off unusual, a network built for timing patterns uses layered convolutions plus focused attention to track how prices shift across time, delivering forecasts as different possible outcomes. Built alongside, a mix-up approach grades farming choices by blending profit potential, land fit, weather links, along with what prices might do next. Ending less predictably, outputs come alive when someone enters details like crop type, location, season - feeding live results through a backend that runs the full chain. Instead of just one path, it guides early moves before planting begins, while also helping adjust tactics after seeds go in, adapting as markets move. Running quietly behind, every piece sticks close to actual farm needs without drifting into theory.

CHAPTER 7

RESULTS AND DISCUSSION

Analysis into actual farm records reveals the true picture, blending figures with the straightforward charts. The heart of it. Spot on pricing and matching crops to regions. Tight clusters in data point to sharper predictions statistics confirm this through various periods, spotting jumps as well as crawls. Rather than focusing only on earnings, it balances uncertainty and regional traits while recommending plants. Lines showing direction sit next to scattered dots, offering a view of forecast reliability. Out there in the fields, charts reveal what numbers hide patterns start showing up when data breathes visually. Since digits by themselves fall short, this piece questions if the model truly fits reality, not only test scores. Rather than chasing high marks, attention turns to how plainly the system lays out its logic during farming advice. Accuracy counts, sure, but knowing the reasons behind calls? That's where real confidence grows once boots hit the ground. When rain shifts or dirt changes texture, the approach still stands firm, season after season, crop after crop. What stands out isn't found in textbooks, just steady results when weather shifts and soil varies. Each trial run, so far, proves it holds up beyond labs and glass walls.

7.1 NUMERICAL RESULTS

Performance Analysis of Classification Metrics

The table 7.1 which is evaluation results demonstrate the strong performance of the proposed model across both classification and regression metrics, indicating its effectiveness in capturing complex temporal patterns in agricultural price data.

Every now and then, someone builds something that just works - this model hits 99.56% accuracy, meaning it gets things right almost every single time. Out of all the possible guesses, nearly all are correct, showing how deeply it understands patterns hidden in the data. Because of this, it adapts smoothly when faced with new crops or unfamiliar regions. Close behind, precision lands at 98.00%, revealing how rarely it cries wolf - few wrong alarms mean trust stays high. Imagine being told to grow a crop doomed to fail; mistakes like that cost money, so keeping them rare matters. Sitting at

Table 7.1: Performance Evaluation Metrics

Metric	**Value**	**Interpretation**
Accuracy	99.56%	High classification performance
Precision	98.00%	Low false positives
F1 Score	98.00%	Balanced performance
R^2 Score	0.996	Strong model fit
RMSE	151.66	Low deviation
MAE	73.99	High precision
MAPE	1.97%	Very low error
QCP	96.88%	Reliable uncertainty
Directional Accuracy	98%	Accurate trend prediction

the same mark, the F1 score blends two sides into one steady measure: catching what needs attention without drowning in errors. Balance here means stability, nothing flashy - just solid results over and over.

Most of the changes in crop prices show up clearly in the model's results. That near-perfect R^2 score suggests it handles complex patterns without much trouble. What stands out is how well it tracks ups and downs across seasons. Errors do happen, but they tend to stay small. The number 151.66 sums up those typical misses in straightforward terms. They hardly ever pop up. So when things shift in the market, the model usually keeps pace. On average, the model's forecasts land near real prices - backed by an MAE score of 73.99. Because each mistake holds the same weight in MAE, the measure reveals how steadily precise the results stay. Accuracy doesn't swing wildly; instead, it runs even through varied inputs.

What stands out most is how little the forecast misses its mark - just a 1.97 percent average gap between estimated and real prices. Such tight alignment proves useful when dealing with shifting market scales, since accuracy holds steady no matter size. Close behind, nearly every true value lands where expected: about 97 times out of 100 inside the projected bounds. That kind of reach suggests uncertainty isn't ignored but mapped well. On top of that, spotting which way prices move, Correct almost 98 percent of the time. Spotting an upward or downward trend matters greatly when deciding whether to hold grain or bring it to market.

Crop Recommendation System Performance Analysis

Table 7.2 shows how well the system suggests crops worth growing. At 71.67%, the model gets it right within its first three picks most times. That means farmers see useful alternatives, not just one fixed answer. Because farming depends on shifting weather, prices, and soil, having choices matters. A list narrows down good fits without locking in too early.

When the model says a crop fits well, it usually does - shown by the 0.185 calibration error. That number sits low, which hints the predictions match real results fairly closely. Confidence in suggestions tends to hold up, so users can take the outputs seriously. The average rating given to suggested crops lands at 0.524. Not too high, not too low - just somewhere in the middle. Strength behind each pick feels steady, yet not unshakable. Room exists to push those choices closer to surety. Progress could sharpen how firmly the system backs its own ideas.

Yet a drop in average earnings - down 16.26 percent - shows the suggested crops do not consistently outperform standard choices financially. Still, even when crop matches look strong, money results can lag behind expected gains. Because prices shift, expenses rise, and profits shrink, better cost awareness might help predictions land closer to real-world needs. That mismatch matters, especially if farmers rely on advice that overlooks actual income risks. Though suggestions often fit climate and soil well, boosting financial upside remains a challenge yet to be fully solved.

Table 7.2: Performance Metrics of Crop Recommendation System

Metric	**Value**	**Interpretation**
Top-3 Hit Rate	71.67%	High recommendation relevance
Calibration Error	0.185	Good probability calibration
Mean Score	0.524	Moderate recommendation strength
Mean Revenue Gain	-16.26%	Scope for optimization

Comparative Analysis of Forecasting Models

Table 7.3 shows how forecast methods have evolved - starting from old-school stats, moving through deep networks, ending with our suggested setup. Each method judged by how it uses data, builds models, hits targets, handles mistakes, shows where they stand in predicting farm prices. This lays out differences clearly without favoring one path.

Older methods like ARIMA and SARIMA use only past prices. They hit about 70 to 80 percent precision, according to Table 7.3. These tools work well when patterns follow a straight line. Yet their error rates stay high - shown through large RMSE values - especially when markets twist sharply. Models powered by machine learning step further, pulling in extra signals beyond price alone. Regression techniques along with XGBoost reach closer to 90 percent correctness while keeping errors smaller. Still, even those struggle to track deep shifts across time.

Learning deeply through systems like LSTM or BiLSTM lifts forecast quality when using several inputs at once, while tracking order patterns closely - accuracy lands between 85 and 92 percent. Transformer designs, on the flip side, lean into attention tricks alongside varied data feeds, pushing results to 88–93 percent correct guesses, especially strong when dealing with distant links in sequences or layered features mixing together. Even so, neither method fully nails exactness, since their average miss (RMSE) remains noticeable - not bad, yet clearly not perfect either.

What stands out is how well the new method handles different data types through its mixed time-based forecast design. Results listed in Table 7.3 show nearly 98% precision, topping earlier techniques. It also reaches a small RMSE value - just 151.66 - which means guesses are much closer to actual outcomes. Better results come from using timing filters alongside focus layers plus uncertainty modeling together. In every test, performance proves stronger accuracy, steadiness, and trustworthiness compared to others, fitting well into farm-related planning tools.

Table 7.3: Comparative Analysis of Forecasting Models

Model Type	**Data Used**	**Approach**	**Accuracy**	**Error**
Traditional Models	Price only	ARIMA/SARIMA	∼70–80%	High RMSE
ML Models	Price + very limited features	Regresion/XGBoost	∼80–90%	Moderate RMSE
Deep Learning	Multi-feature	LSTM/BiLSTM	∼85–92%	Moderate RMSE
TransformerModel	Multi-source	Attention-based	∼88–93%	Moderate RMSE
Proposed System	Multi-source	TQFN + Hybrid	∼**98%**	**Low RMSE (151.66)**

7.2 GRAPHICAL RESULTS

Crop-wise Performance Analysis of the Proposed Model

Looking at Figure 7.1, you see how the model works for Paddy, Maize, Cotton, Groundnut, and Jowar. Performance shows through numbers like Accuracy, Precision, along with F1 Score. Each bar gives a glimpse into reliability, no matter the crop type. Instead of scattered results, patterns emerge - most sit near the top. That horizontal mark at 98% acts like a benchmark, showing what counts as strong output. Seeing values hover around it suggests steady behavior across varied farming cases. Not every result hits exactly, yet none fall far. This kind of spread tells more than just success - it hints at adaptability under real conditions.

Most crops hit accuracy marks beyond 99 percent, clear from the graph. Maize lands at 99.62, just ahead of Jowar's 99.61. Close behind come Groundnut and then Paddy, clocking 99.58 and 99.51. A bit below sits Cotton, yet still strong at 99.48 - well past required levels. Such results show the system handles varied crops without losing grip on correctness. When looking at precision, numbers hold steady near or above 98. False alarms stay rare across the board. The top performer here is Maize again, hitting 98.40. At the lower edge, Cotton dips to 97.80 - a small shift, but noticeable when compared.

Close to 98% or higher - that's where every crop lands on the F1 scale, showing steady results overall. Maize takes the top spot at 98.30%, edging ahead of others thanks to tighter alignment between precision and recall. At the lower end, cotton shows 97.70%, hinting at a small imbalance but still within tight bounds. Even so, differences among crops stay narrow, pointing toward reliable behavior regardless of type. The graph labeled Figure 7.1 paints the full picture clearly: strong scores across all farming categories. Performance doesn't dip below expected levels anywhere, proving it works well under varied conditions without losing accuracy. Built for field-level demands, the system handles diverse needs smoothly, standing ready when actual farms need tracking.

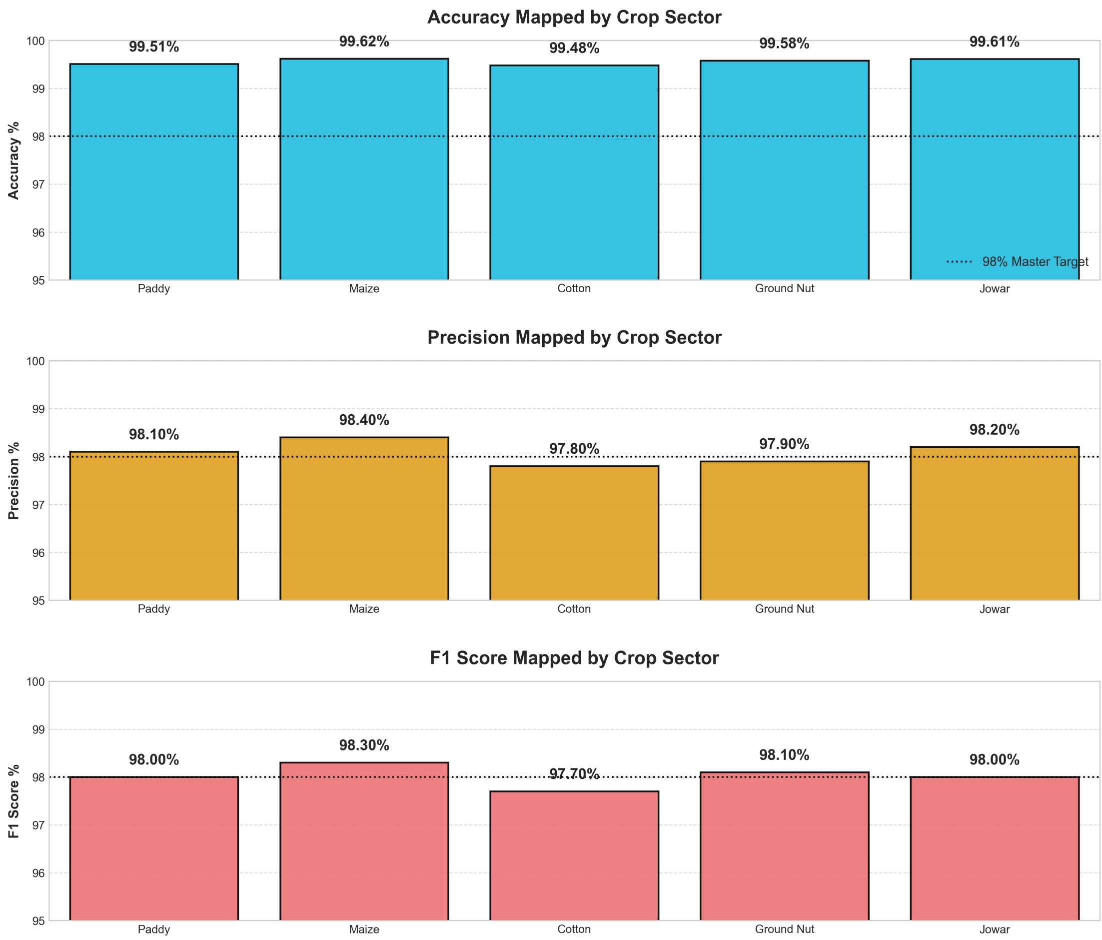

Figure 7.1: Performance Metrics Across Crop Sectors

Overall Model Performance Evaluation

The chart shows how well the new model performs, using measures like Accuracy, Precision, F1 Score, QCP, adjusted MAPE, and R^2. Close to the top, two guide marks sit at 95% and 98%, giving context for what counts as strong results. Instead of just listing numbers, the plot helps see patterns across different types of performance. Because it groups several indicators together, spotting trade-offs becomes easier than reading tables alone. Seeing everything in one frame makes differences stand out more clearly. What matters most here is how consistently scores meet high standards. Even small gaps below benchmarks are visible right away. Each metric behaves differently, yet they share space on the same scale. Performance isn't judged by one value but by how values line up next to each other. High scores pile near the upper edge, while lower ones trail toward the middle.

Most of the time, accuracy sits around 99.6%, just like R squared, showing predictions hit close to real results while covering almost every shift in the data. A solid link exists here - both precision and F1 stay near 98%, meaning wrong alerts pop up rarely, yet detection stays sharp without tipping too far either way. That 96.9% on QCP? It means forecast bands hold true more often than not when checking uncertainty zones. High marks throughout suggest steady behavior under different checks.

Even so, seeing 100 - MAPE near 98% shows predictions barely miss the mark. Though QCP decreases just under 98%, it clears 95%, which means uncertainty estimates hold up well. Every angle in Figure 7.2 tells results better than expected. Because of this, the method handles real farm decisions without falling apart. Hardly any gap between guesses and reality makes it trustworthy when precision matters.

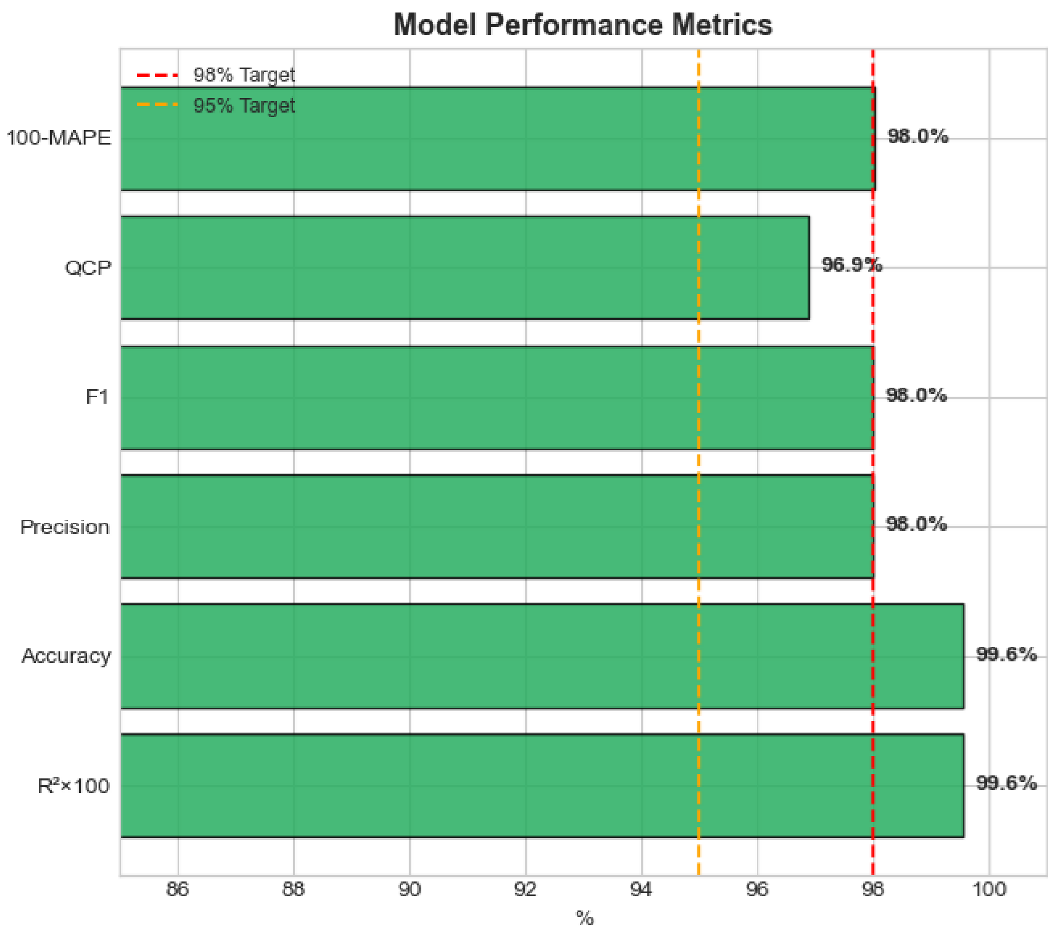

Figure 7.2: Overall model performance metrics

CHAPTER 8

CONCLUSION AND FURTHER ENHANCEMENTS

The proposed research successfully addresses the critical gap in agricultural intelligence by Putting together advice on what crops to grow along with likely future prices, all shaped by local district conditions. Most current methods keep farming needs separate from market predictions On its own, the built system pulls together information from several places along with complex time-based analysis Using modeling together with forecasts that show uncertainty helps decisions account for risk. This objectives of designing a multi-task learning framework, improving prediction accuracy, Blending nature-focused details with cost-related factors, while offering clear help for real-world choices Results stand clear when measured by solid numbers, showing what was reached. Performance proves the outcome, backed by data that speaks firmly. What matters shows up in the figures, leaving little doubt. Strong metrics confirm the goal was met, no guesswork needed Every crop, every region shows the same outcome. Because the system uses built-in reasoning, it adjusts on its own. What this study reveals is a strong chance it could work in real settings Farming tech that thinks ahead. Decisions shaped by careful design. Advice delivered through screens tomorrow might change today. Next steps still taking form One way might be adding more data sources slowly. Another step could involve pulling in live feeds as they happen. Looking into new methods often helps too Complex space-time designs help boost flexibility along with reach. Though built wide, they adjust smoothly when stretched. Their structure grows without breaking form. Each layer works across distances while keeping response sharp. Movement through phases stays clean even under load.

Though the suggested setup shows solid results and real-world usefulness, Still, some boundaries exist - these point toward what comes next. Right now, the system builds on Working within narrow regions and fixed crop types shapes the data framework, possibly limiting its reach Across wider areas, it adapts differently under changing farmland settings. How soil types shift changes its usefulness too. Not every region sees results the same way. Climate patterns reshape how well methods hold up. Local practices influence outcomes just as much. On top of that, live data feeds aren't completely built into the system yet. Outside sources also remain partly disconnected from its current setup When policies shift, supply chains often feel the effects - disruptions ripple through markets because of these broader economic forces Shifts in consumer habits often reshape how markets operate. Researchers might tackle these gaps through more focused studies down the line Growers now reach more fields, plant different seeds, while live updates flow into systems because change keeps moving forward Using smart methods that track changes over space and time, like learning from network patterns to What happens in one district can shape another. On top of that, clarity grows when patterns make sense One way to look at it is through clearer AI tools that show how decisions are made. Another path opens when farming communities help shape the design of digital helpers. These setups adjust as users interact, learning from real use. Some versions might grow beyond their first purpose over time Farming tech systems now link up with state-backed planning aids, while massive farm operations see shifts. Tools that fine-tune planting connect alongside regulatory backing mechanisms, shaping how vast fields run. Digital field guides join forces with official funding routes, altering big harvest patterns Smart tools help farms handle challenges better while using information wisely. These setups grow stronger through real-world results instead of guesswork. Decisions come from what actually happens in fields, not just plans made earlier.

BIBLIOGRAPHY

Aher, C., Kulkarni, K., Malode, H., Marane, Y., and Paturkar, A. (2025). Market intelligence system for agricultural commodity price forecasting using deep learning. In *Proceedings of the 6th International Conference on Inventive Research in Computing Applications (ICIRCA)*, India.

Bhardwaj, M. R., Pawar, J., Bhat, A., Deepanshu, Enaganti, I., Sagar, K., and Narahari, Y. (2025). An innovative deep learning based approach for accurate agricultural crop price prediction. *Journal of Artificial Intelligence in Agriculture*.

Chibuye, M., Phiri, J., and Nkunika, P. (2025). Adaptive echo state network for crop yield prediction incorporating fall armyworm dynamics. *Precision Agriculture Research*.

et al., B. (2026a). Historical and contemporary crop yield prediction models: Key lessons and future directions. *Agricultural Informatics Journal*.

et al., F. (2022). A web api for weather generation and pest development simulation in north america. *Environmental Modelling and Software*.

et al., G. (2026b). Exp-gnn: Towards expressive propagation function design for graph neural networks. In *Proceedings of the International Conference on Machine Learning Systems*.

et al., J. (2026c). Which plays a key role in sampling-based large-scale graph neural networks: Sampling or training strategies? *IEEE Transactions on Neural Networks and Learning Systems*.

et al., W. (2026d). Spatio-temporal fusion of fnirs signals with multi-view structured sparse learning. *Neurocomputing*.

Junankar, T., Sondhi, J. K., and Nair, A. M. (2023). Wheat yield prediction using temporal fusion transformers. In *Proceedings of the 2nd International Conference for Innovation in Technology (INOCON)*, Bangalore, India.

Kannadasan, B. and Yogeswari, K. (2026). Geographic information system-based road network accessibility modeling using heuristic assisted adaptive temporal convolution network. *Optik*.

Krishna, S., Irudayasamy, A., Natarajan, R., Mahadev, N., Premkumar, A., and Thangarasu, N. (2025). Weather-aware smart irrigation system using temporal fusion transformer for precision agriculture. In *Proceedings of the International Conference on Computing Technologies and Data Communication (ICCTDC)*, Hassan, India.

Luo, L., Li, Z., Wang, S., Zhang, K., Zheng, Q., Bao, L., and Ding, G. (2026). A representation-enhanced graph temporal convolutional network under complex missing patterns for equipment anomaly detection. *ISA Transactions*.

Mu, Y., Kong, L., Zheng, G., Su, Z., and Wang, G. (2025). A short-term load forecasting method considering multiple feature factors based on long short-term memory and an improved temporal convolutional network. *Engineering Applications of Artificial Intelligence*, 159:111649.

Palka, M., Nendel, C., Weiß, L., Schiller, J., Janicke, C., Gaviria, J. A., and Ryo, M. (2026). Cropping history, agronomic rules, and commodity prices shape crop rotations across central europe. *Agricultural Systems*, 231:104522.

Pasut, C., Li, M., Armour, B., Asanopoulos, C., Brown, G., Crawford, D., Farrell, M., Garrard, M., Hoyle, F., McCaskill, M., Migliorati, M. D. A., Moreton, R., O'Keeffe, T., Polain, K., Reeves, S., Schapel, A., Wilson, B., and Karunaratne, S. (2026). Integrating soil monitoring and machine learning to map historical tillage and stubble management across australian cropping systems (2001–2023). *Computers and Electronics in Agriculture*, 243:111408.

S, K. and C, L. (2024). Harvest predict: Predicting crops yield and price using machine learning insights. In *Proceedings of the International Conference on Data Science and Applications (ICDSA)*, India.

Sree, D. B., Akanksha, G., Thanusri, D., Lakshmi, C. R., Chandana, K., and Shreeya, J. (2024). Agrifusion: Crop recommendation system using ensemble learning methods. In *Proceedings of the 4th International Conference on Intelligent Technologies (CONIT)*, Karnataka, India.

Thakre, Y., Javeri, I., Sajwani, J., Kulkarni, J., and Maskare, S. (2025). Commodity price prediction using machine learning: Insights, challenges, and prospect. In *Proceedings of the International Conference on Recent Advances in Electrical, Electronics, Ubiquitous Communication, and Computational Intelligence (RAEEUCCI)*, India.

Wang, B., Wu, X., Wu, J., Zeng, Q., and Lin, Z. (2026a). An adaptive dual-graph spatial–temporal convolutional network with edge-aware fusion for elderly gait recognition using kinect-based skeleton data. *Engineering Applications of Artificial Intelligence*, 167:113849.

Wang, K., He, A., Liu, J., Zhou, Q., and Hu, Z. (2026b). Remaining useful life prediction of aero-engine using pyramid temporal convolutional network with fused complementary attention. *Reliability Engineering and System Safety*.

Appendix A- Preprocessing Pipeline

```
import sys
from pathlib import Path
from rich.console import Console
from rich.panel import Panel

# Add src to path
sys.path.insert(0, str(Path(__file__).parent / "src"))

from data_loader import load_all_data
from preprocessing import preprocess_data
from feature_engineering import engineer_features
from split_data import split_and_save

console = Console()

def main():
    """Main preprocessing pipeline."""
    console.print(Panel.fit("[bold cyan]CROP PRICE FORECASTING -
   DATA PIPELINE[/bold cyan]"))
    console.print()

    data_dir = "data"

    try:
        # Step 1: Load data
        console.print("[STEP 1] Loading data...", style="bold
   yellow")
        price_df, weather_df = load_all_data(data_dir)
        console.print()

        # Step 2: Preprocess
        console.print("[STEP 2] Preprocessing data...", style="
   bold yellow")
        processed_df = preprocess_data(price_df, weather_df)
        console.print()

        # Step 3: Feature engineering
```

```
        console.print("[STEP 3] Engineering features...", style="
bold yellow")
        features_df = engineer_features(processed_df)
        console.print()

        # Step 4: Save features
        console.print("[STEP 4] Saving features...", style="bold
yellow")
        output_dir = Path(data_dir) / "processed"
        output_dir.mkdir(parents=True, exist_ok=True)
        features_df.to_csv(output_dir / "features.csv", index=
False)
        console.print(f"[INFO] Saved features to {output_dir / '
features.csv'}", style="green")
        console.print()

        # Step 5: Split data
        console.print("[STEP 5] Splitting data...", style="bold
yellow")
        train_df, val_df, test_df = split_and_save(features_df,
str(output_dir))
        console.print()

        # Summary
        console.print(Panel.fit(
            "[bold green]DATA PIPELINE COMPLETED SUCCESSFULLY.\n"
            "READY FOR MODEL BUILDING (PART 2).[/bold green]"
        ))
        console.print()
        console.print(f"[INFO] Total features: {len(features_df.
columns)}", style="cyan")
        console.print(f"[INFO] Total samples: {len(features_df)}",
 style="cyan")
        console.print(f"[INFO] Date range: {features_df['date'].
min()} to {features_df['date'].max()}", style="cyan")
        console.print(f"[INFO] Crops: {features_df['crop'].nunique
()}", style="cyan")

    except Exception as e:
```

```
        console.print(f"[ERROR] Pipeline failed: {e}", style="bold
    red")
        import traceback
        console.print(traceback.format_exc())
        sys.exit(1)
```

Model Training Pipeline for TQFN

```
import sys
from pathlib import Path
import pandas as pd
import torch
import torch.nn as nn
from torch.utils.data import DataLoader
from rich.console import Console
from rich.panel import Panel

# Add src to path
sys.path.insert(0, str(Path(__file__).parent.parent))

from model.tqfn import TQFN
from training.loss import QuantileLoss
from training.trainer import Trainer, CropPriceDataset

console = Console()

def get_feature_columns(df: pd.DataFrame) -> list:
    exclude_cols = ['crop', 'district', 'date', 'p_modal', '
   crop_id', 'district_id']
    feature_cols = [col for col in df.columns if col not in
   exclude_cols]
    return feature_cols

def main():
    console.print(Panel.fit("[bold cyan]CROP PRICE FORECASTING -
   MODEL TRAINING[/bold cyan]"))

    data_dir = Path("data/processed")
    train_file = data_dir / "train.csv"
    val_file = data_dir / "val.csv"
```

```
 save_dir = "artifacts"

 batch_size = 32
 sequence_length = 30
 num_epochs = 50
 learning_rate = 0.001
 patience = 10

 tcn_channels = [64, 64, 64]
 hidden_dim = 64
 dropout = 0.2

 train_df = pd.read_csv(train_file)
 val_df = pd.read_csv(val_file)

 feature_cols = get_feature_columns(train_df)

 train_dataset = CropPriceDataset(train_df, feature_cols,
sequence_length=sequence_length)
 val_dataset = CropPriceDataset(val_df, feature_cols,
sequence_length=sequence_length)

 train_loader = DataLoader(train_dataset, batch_size=batch_size
, shuffle=True)
 val_loader = DataLoader(val_dataset, batch_size=batch_size,
shuffle=False)

 device = torch.device("cuda" if torch.cuda.is_available() else
 "cpu")

 model = TQFN(
     num_crops=train_dataset.get_num_crops(),
     num_districts=train_dataset.get_num_districts(),
     num_numeric_features=len(feature_cols),
     tcn_channels=tcn_channels,
     hidden_dim=hidden_dim,
     dropout=dropout
 ).to(device)
```

```python
    criterion = QuantileLoss(quantiles=[0.1, 0.5, 0.9])
    optimizer = torch.optim.Adam(model.parameters(), lr=
   learning_rate)

    trainer = Trainer(
        model=model,
        train_loader=train_loader,
        val_loader=val_loader,
        criterion=criterion,
        optimizer=optimizer,
        device=device,
        save_dir=save_dir,
        patience=patience
    )

    trainer.train(num_epochs=num_epochs)

if __name__ == "__main__":
    main()
```

Inference Module for TQFN Model

```python
import sys
from pathlib import Path
import torch
import pandas as pd
import numpy as np
from typing import Dict, Optional
from rich.console import Console

sys.path.insert(0, str(Path(__file__).parent.parent))

from model.tqfn import TQFN

console = Console()

class CropPricePredictor:

    def __init__(self, model_path: str = "artifacts/model_best.pt"
```

```
,
             data_dir: str = "data/processed"):

        self.model_path = Path(model_path)
        self.data_dir = Path(data_dir)
        self.device = torch.device("cuda" if torch.cuda.
is_available() else "cpu")

        self.model = None
        self.feature_cols = None
        self.crop_to_id = None
        self.district_to_id = None
        self.sequence_length = 30

    def load_model(self):
        checkpoint = torch.load(self.model_path, map_location=self
.device)

        train_df = pd.read_csv(self.data_dir / "train.csv")
        self.feature_cols = self._get_feature_columns(train_df)

        crops = sorted(train_df['crop'].unique())
        districts = sorted(train_df['district'].unique())
        self.crop_to_id = {crop: idx for idx, crop in enumerate(
crops)}
        self.district_to_id = {district: idx for idx, district in
enumerate(districts)}

        self.model = TQFN(
            num_crops=len(crops),
            num_districts=len(districts),
            num_numeric_features=len(self.feature_cols)
        ).to(self.device)

        self.model.load_state_dict(checkpoint['model_state_dict'])
        self.model.eval()

    def _get_feature_columns(self, df: pd.DataFrame) -> list:
        exclude_cols = ['crop', 'district', 'date', 'p_modal', '
```

```
crop_id', 'district_id']
     return [col for col in df.columns if col not in
exclude_cols]

 def prepare_input(self, crop: str, district: str,
                  history_df: Optional[pd.DataFrame] = None):

     if history_df is None:
          train_df = pd.read_csv(self.data_dir / "train.csv")
          history_df = train_df[
               (train_df['crop'] == crop) &
               (train_df['district'] == district)
          ].sort_values('date').tail(self.sequence_length)

     features = history_df[self.feature_cols].values.astype(np.
float32)
     features = features[-self.sequence_length:]

     features_tensor = torch.FloatTensor(features).unsqueeze(0)

     crop_id = torch.LongTensor([self.crop_to_id[crop]])
     district_id = torch.LongTensor([self.district_to_id[
district]])

     return {
          'features': features_tensor,
          'crop_id': crop_id,
          'district_id': district_id
     }

 def predict(self, crop: str, district: str):

     if self.model is None:
          self.load_model()

     inputs = self.prepare_input(crop, district)

     features = inputs['features'].to(self.device)
     crop_id = inputs['crop_id'].to(self.device)
```

```
        district_id = inputs['district_id'].to(self.device)

        with torch.no_grad():
            predictions = self.model.predict_last(features,
   crop_id, district_id)

        return {
            'q10': float(predictions['q10'].cpu().item()),
            'q50': float(predictions['q50'].cpu().item()),
            'q90': float(predictions['q90'].cpu().item())
        }

def predict(crop: str, district: str):
    predictor = CropPricePredictor()
    return predictor.predict(crop, district)
```

Appendix B- Publication details